

PROJECT REPORT

Title: Attendance scenario calculator

Course Name: Python Essentials

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Submitted By: RAJ GUPTA

Registration/Student ID: 26BCY10108

1.Introduction

Attendance is an important part of academic planning because students often need to know their current attendance percentage, the number of classes they may miss, or the number of additional classes they must attend to reach a target percentage.

The Attendance Scenario Calculator is a Python-based command-line project designed to make these calculations quick, clear, and understand. The application accepts basic attendance information such as total classes conducted and classes attended.

It calculates the current attendance percentage and provides scenario-based results for future attendance decisions. For example, it can estimate how many upcoming classes a student needs to attend continuously to reach a target percentage, or how the percentage may change after attending or missing a specified number of classes.

The project demonstrates core Python Essentials concepts including variables, input and output, arithmetic operations, conditional statements, loops, functions, dictionaries, validation, exception handling, modular programming, and formatted output. It is intended as an educational project and does not connect to a university attendance database or make official academic decisions.

2. Problem Statement

Students commonly calculate attendance manually using formulas and repeated trial-and-error. This can lead to calculation mistakes and makes it difficult to answer practical questions such as: 'How many classes do I need to attend to reach 75%', 'What happens if I miss two more classes?', or 'What will my attendance be after attending the next ten classes?' The project solves this problem through a simple scenario calculator.

3. Objectives

1. Calculate current attendance percentage accurately.
2. Estimate classes needed to reach a target attendance Percentage.
3. Predict attendance percentage after future attended or missed classes.
4. Validate user input and prevent invalid calculations.
5. Provide simple, readable terminal output suitable for a beginner.

4. Functional Requirements

FR1---- Accept total classes conducted and classes attended.

FR2 ----Calculate current attendance percentage.

FR3 ---- Calculate classes needed to reach a target percentage.

FR4 ----Calculate future attendance after upcoming attended classes.

FR5 ---- Calculate the effect of upcoming missed classes.

FR6 ----Display clear results in the terminal

5.Non-Functional Requirements

1.Usability: The program should be simple enough for a beginner to operate from a terminal.

2. Reliability: Invalid, negative, or inconsistent inputs should be rejected with a clear message.

3. Maintainability: Calculation and validation logic should be separated into modules/functions.

4. Portability: The project should run with Python and the standard library without a GUI dependency.

6. Major Project Module

1.Input & Validation ----- Reads attendance values and checks whether they are valid.

2. Attendance Calculator----- Computes current and future attendance percentages.

3.Scenario Engine ----- Determines classes required or the effect of future attendance changes.

3.Report/Output --- Presents results in a clear terminal format.

4.Report/Output Presents results in a clear terminal format.

7. Technology and Python Concepts Used

1. Python 3
2. Functions and modular programming.
3. Conditional statements and loops.
4. Lists/dictionaries for structured Information.
5. Exception handling for invalid input.
6. Formatted string output.
7. Command-line execution.

8. Attendance Calculation Logic

Current Attendance: $\text{Attendance \%} = (\text{Classes Attended} / \text{Total Classes Conducted}) \times 100$

Future Attendance: $\text{Future \%} = ((\text{Attended} + \text{Future Attended}) / (\text{Total} + \text{Future Attended} + \text{Future Missed})) \times 100.$

Target Scenario: The program increases the number of future attended classes until the calculated percentage reaches the user-selected target, while showing the required count.

9.Example Scenario

Total classes conducted	=	60
Classes attended	=	42
Current attendance	=	70%
Target attendance	=	75%
Upcoming missed classes	=	0

The calculator can then show how many consecutive classes must be attended to reach the target. The result is intended for planning only; actual university rules may use different policies, rounding methods, or attendance exclusions.

10. System Workflow

- 1. Start the program.**
- 2. Enter total classes conducted.**
- 3. Enter classes attended.**
- 4. Validate the entered values.**
- 5. Calculate current attendance.**
- 6. Select a scenario: reach target / future attendance / missed classes.**
- 7. Perform calculation.**
- 8. Display the result and end.**

11. Use Case Description

Primary Actor: Student/User

1. Enter attendance details.
2. View current attendance percentage.
3. Select an attendance scenario.
4. View the calculated result.
5. Repeat with different scenarios

12. Sample Terminal Output

ATTENDANCE SCENARIO CALCULATOR

1. Calculate current attendance
2. Find classes needed for target
3. Predict future attendance
4. Exit

Enter choice: 2

Total classes: 60

Classes attended: 42

Target percentage: 75

Current attendance: 70.00%

Classes required to reach target: 12

13. Error Handling and Validation

1. Total classes must be greater than zero.
2. Attended classes cannot be negative.
3. Attended classes cannot be greater than total classes.
4. Target attendance should be between 0% and 100%.
5. Scenario counts such as future attended/missed classes should not be negative.
6. Invalid text input should be handled without crashing the program.

14. Testing Plan

test case	expected result
Valid attendance: 40/50	80% displayed
Attended > Total	Validation error
Negative value	Validation error
Target below current	No unnecessary classes required
Future attended classes	New percentage calculated
Future missed classes	Reduced percentage calculated

15. Challenges Faced

- 1.Designing formulas for different attendance scenarios.
2. Avoiding division-by-zero and impossible input combinations.
- 3.Making the terminal interaction simple and readable.
- 4.Separating program logic into reusable modules.
- 5.Applied Python fundamentals to a practical problem.

16. Learning Outcomes

1. Applied Python fundamentals to a practical problem.
2. Improved understanding of functions, validation, and modular design.
3. Learned to translate a real-world scenario into formulas and algorithms.
4. Practiced command-line execution and software testing

17. Future Enhancements

1. Add a graphical user interface.
2. Store student/subject attendance records on a database.
3. Support multiple subjects and monthly attendance history.
4. Add charts and visual attendance trends.
5. Export reports to CSV or PDF.
6. Add configurable institutional attendance rules

18. Limitations

- 1.The project calculates attendance from user-entered values.
- 2.It does not access official college attendance records.
- 3.The result should not be treated as an official academic decision.
- 4.The calculations depend on the data entered by the user.

19. Conclusion

The Attendance Scenario Calculator is a practical Python Essentials project that converts a common student problem into a simple command-line application. By combining input validation, arithmetic logic, scenario analysis, and modular programming, the project demonstrates how Python can be used to build a useful real-world utility. The design can also be extended into a larger attendance management system in the future.

20. References

1. Python 3 documentation - basic syntax, functions, exceptions, and standard library.
2. Course learning material for Python Essentials.
3. Project formulas derived from standard attendance percentage calculations.

My Declaration: This report describes an educational project prepared for the Python Essentials course.

